Equity Data Audit: Cleaning, Adjustment Status, and Cross-Validation

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> *Covers 20 years of cleaned CRSP daily stock data (1995-2014), the split/dividend adjustment-status finding, and a cross-check against a second price file -- the output of Phase 1, Slice 2 of this project.*

What this dataset is, and why it needed cleaning

This report covers the first stage of building this project's stock-price history: cleaning a raw pull from CRSP (Center for Research in Security Prices), a widely used academic database of US daily stock prices and trading data.

The cleaned result covers 20 years (1995-2014), 20,478,337 (stock, day) rows across 8,100 distinct stocks. Each stock has a permanent numeric ID (a "permno") that never changes or gets reused, unlike its ticker symbol, which a company can change or which can be reassigned to a different company later.

Two of CRSP's own data conventions look like errors if you don't know about them, so this cleaning step checked for both directly rather than assuming they don't apply here. First, when a stock doesn't trade on a given day, CRSP records an estimated price with a minus sign in front of it as a marker that it isn't a real trade price; 110,227 rows (0.54% of rows that have a price at all) fall into this case, and the minus sign is removed so the price is usable, with a separate flag column kept so this is never silently hidden. Second, CRSP documents four special codes that can appear in place of a return, in either the `ret` or `retx` column, to mean "we don't have this," not a literal huge gain or loss; this cleaned data has 0 rows with one of those codes remaining in either column.

Are prices and returns already adjusted for stock splits and dividends?

This question matters a lot for anything built on top of this data. A stock split (for example, "2-for-1") doubles the number of shares and roughly halves the price without changing what the company is actually worth or what an investor actually earned. A dividend is a cash payment to shareholders, which is part of an investor's real return even though it doesn't show up as a change in the share price. If a price series isn't adjusted for these events, a split looks exactly like a crash, and a return series that ignores dividends understates how much money was actually made.

Three things are checked directly against this pull's own numbers, across every year in the dataset -- not assumed, and not standing in for each other. Splits and distributions: 5,363 adjustment events had a raw price move bigger than 20% (consistent with a real split or distribution, not day-to-day noise); on 0.02% of those events, CRSP's actual reported return still matched that same raw, unadjusted move -- a fully unadjusted return column would measure close to 100% here. Dividends: of 20,244,672 rows with both `ret` and `retx` populated, they disagree on 135,756 (0.67%) -- consistent with real, sporadic dividend payments, not a mislabeled column -- and only 0 of those disagreements run in the direction a real dividend (a cash inflow) essentially never should. Return scale: across 20,244,672 non-null `ret` values, only 77,789 (0.38%) exceed 20% in size -- a small share is what a correctly-scaled fractional daily return column (e.g. 0.05 meaning 5%) should look like; a column mistakenly left in percent units (e.g. 5 meaning 5%) would instead put nearly all of its values above that same threshold.

Chart 1 shows one specific, named example ON TOP OF these population-level figures, not the basis for the verdict above: a specific stock's actual traded price around one of its stock splits, next to what an investor's money would have looked like over that same stretch if you build it up purely from the day-by-day return column instead of the price column. If the return column already accounts for the split, the price line should show an abrupt, split-sized drop on the split date while the return-built line shows no such split-sized break -- just the ordinary day's return, the same size as any other day's -- and that is exactly what Chart 1 shows.

Why this data has no split-adjusted price column

CRSP also provides `cfacpr`, a cumulative split-adjustment factor that could, in principle, be divided into the raw price column (`prc`) to get a split-adjusted price level. This report deliberately does not do that, and does not provide such a column: `cfacpr` is cumulative to the LAST date in whatever file this project happened to pull, not to the end of this project's own sample window. Dividing an early row's price by that same `cfacpr` would silently bake in every split that happened between that row's date and the pull's own end date -- splits that hadn't happened yet, and couldn't have been known, at that earlier point in history. That is exactly the kind of look-ahead mistake this project has to avoid, and it holds regardless of the adjustment-status finding above: `cfacpr`'s own definition is the problem, not whether `ret` happens to already be adjusted. Plainly: `cfacpr` must not be used as a divisor.

Concretely: of 8,100 stocks with data anywhere in this window, 997 end the window with `cfacpr` different from 1 -- meaning a future re-pull (picking up whatever splits happen between now and then) would silently change any derived adjusted price for rows that are supposed to be permanent history. Applying a simple "at least $5" price screen to the first year's data: using the actual traded price versus using `prc / cfacpr` for the same rows, 106,187 rows would be wrongly excluded and 21,443 would be wrongly included -- 864 different stocks affected. One more real caveat on top of the look-ahead one above: 15,121 rows have `cfacpr` exactly equal to 0, where `prc / cfacpr` is mathematically undefined (a division by zero), not just look-ahead-tainted -- another reason not to divide by it without guarding first. `prc` and `cfacpr` are both still in the output for anyone who explicitly wants `prc / cfacpr` with these caveats in view; for

comparing returns across a split safely, use `ret`/`retx` instead (see the evidence above), which don't have this particular problem.

Cross-checking against a second price file

This project separately had access to an older, non-CRSP daily price file (one row per ticker per day) that predates its use of CRSP. Comparing that file against CRSP is a useful cross-check: if two differently-sourced price series agree closely, that's good evidence neither one has a scaling or adjustment mistake baked in. The two files only overlap for a few of this project's years, so this check can only speak to that overlapping stretch, not the full window this project ultimately covers.

The second file has real data-quality problems of its own, disclosed here rather than glossed over: it has 48,633 rows with a genuinely empty price field, 3,660 with an infinite (`inf`/`-inf`) price, and 27,924 with a negative price with no documented convention behind it (unlike CRSP's own bid/ask-average convention) (counted from 2011-01-03 to 2013-12-31, not the whole file, which runs years further) -- all treated as invalid and left blank rather than guessed at. Ticker-date price collisions (more than one price recorded for the same ticker on the same day) drop a further 61,343 rows on the second file's side and 5,284 rows on CRSP's side. Only 35.86% of CRSP's 4,947 in-window tickers found any match at all in the second file -- the second file only contains tickers that still existed when it was downloaded, so this whole check skews toward companies that survived, and says nothing about ones that didn't.

Chart 2, below, compares `retx` (CRSP's price-only return) rather than `ret` (established above as the dividend-adjusted column): the second file's own prices show no sign of dividend reinvestment, so `retx` is the honest like-for-like comparison -- using `ret` instead would make the disagreement look much larger specifically on ex-dividend days (on this run's 10,867 ex-dividend days for retained segments, median absolute difference against the second file is 6.18e-03 using `ret` versus 2.46e-07 using `retx`). Worth flagging: the two files agree to far more decimal places than two genuinely independent, unrelated vendors plausibly would on separate measurements of the same real-world price. That's better evidence that the two files apply the same underlying convention than that two unrelated measurements of reality agree -- so this should not be read as full independent confirmation. A point sitting exactly on Chart 2's dashed diagonal line means the two sources agree perfectly on that day; the closer the whole cloud hugs that line, the stronger the agreement.

Coverage: how much data do we actually have?

Chart 3 shows, for every year in the cleaned dataset, how many (stock, day) rows survived cleaning and how many distinct stocks contributed at least one of them. 2008 has the most rows of any year, 1,116,862; across all years, rows per year range from 516,771 to 1,116,862, and distinct stocks per year range from 3,581 to 4,731. 2014's bar only covers data through 2014-06-30, not the full calendar year, so it isn't directly comparable to the full-year bars beside it.

What this means for the project

None of the checks in this report turned up a case of unadjusted returns, a missing dividend adjustment, or an implausible return scale -- this is CRSP's standard, well-documented convention, confirmed directly against this project's own pull rather than assumed. That means the cleaned return columns in data/normalized_equity/ are safe to use as-is for the return calculations later stages of this project depend on. The cross-check against a second price file, while it can only speak to a few of this project's years, adds further confidence where it applies: the two sources agree closely almost everywhere they overlap -- though see the caveat above about just how closely, which is itself informative about how independent the two files really are.

Three real limitations carry forward from here, not one: coverage (which years and stocks are actually present, shown in Chart 3); delisting returns (a stock that gets delisted mid-study simply stops appearing in the cleaned data on its last trading day -- no final delisting return is captured, which is explicitly out of scope for this slice); and the `cfacpr` look-ahead trap described above (do not use `prc / cfacpr` as a divisor).

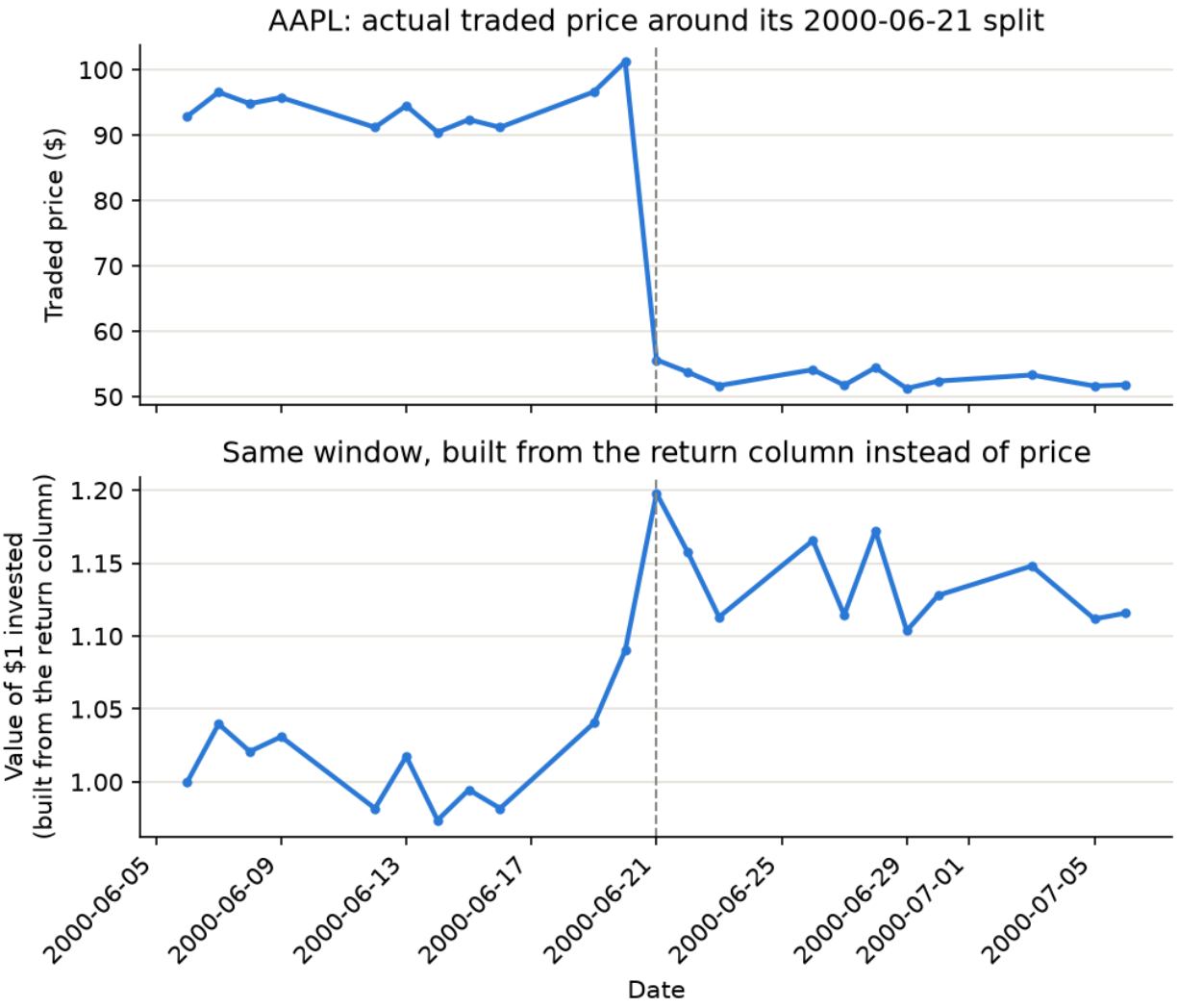


Chart 1. AAPL's actual traded price fell 45.06% overnight on 2000-06-21 (from $101.25 to $55.62), while the return column recorded +9.88% that same day -- a much smaller-magnitude number reflecting the stock's real return, not the raw price change. The stock split accounts for the gap between the two: reading the price column alone would look like a one-day crash that didn't really happen. The bottom panel's return-built value of $1 invested has no split-sized break in it -- it still steps by that same ordinary +9.88%, it just isn't distorted by the split itself the way the price panel is.

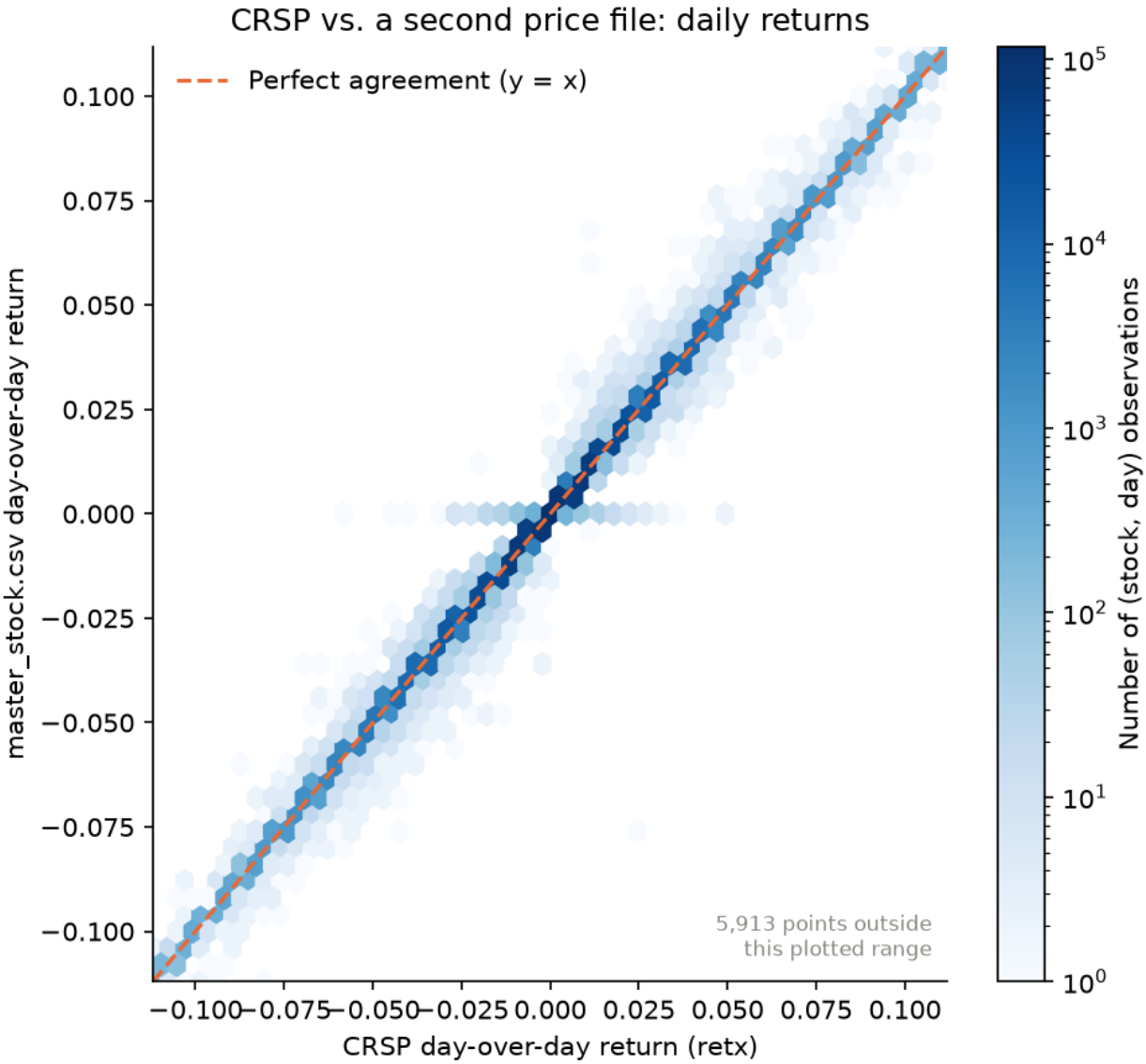


Chart 2. Each hexagon groups together many (stock, day) observations, with a return recorded in both CRSP and the second price file, between 2011-01-03 and 2013-12-31, colored by how many observations fall in it (darker/more intense = more) -- 1,226,003 stock-days plotted; 5,913 of the largest disagreements fall outside this plotted range and aren't shown. The two sources agree closely overall: a correlation of 0.9810, and 99.09% of stock-days agreeing to within 1 percentage point. Setting aside 36 segment(s) (19,002 rows) where the two files appear to track different companies under a reused ticker symbol; and 5 segment(s) (759 rows) with too few usable overlapping days to judge either way, agreement rises to a correlation of 0.9998 and 99.94% within 1 percentage point.

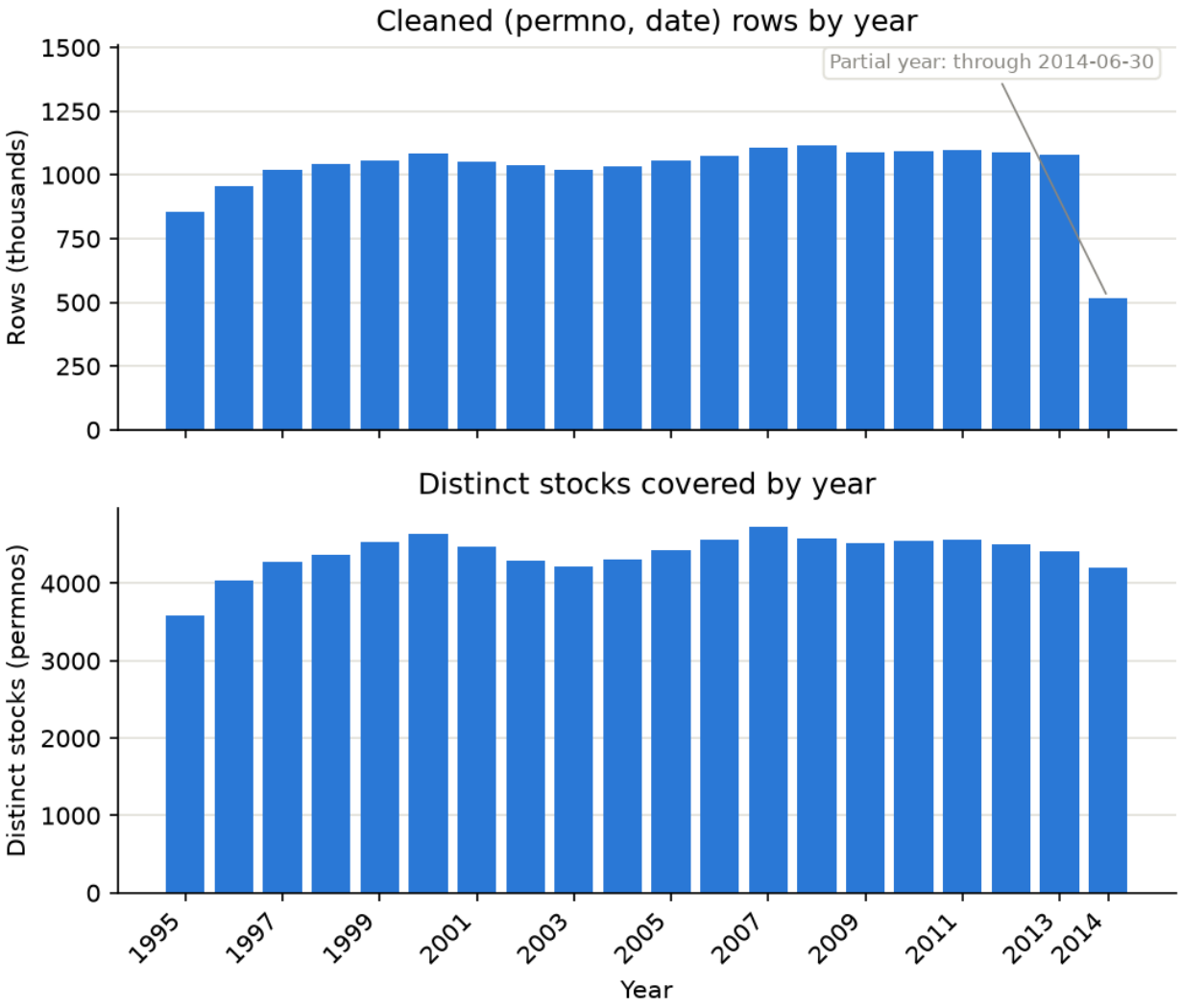


Chart 3. Rows and distinct stocks covered per year, 1995-2014, across 20,478,337 total rows and 8,100 distinct stocks.